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FOOD DELIVERY MANAGEMENT SYSTEM

DBMS ASSIGNMENT

Customers , Restaurants, Orders ,
Payments, Delivery



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NAME AND PURPOSE OF THE APPLICATION

Application Name:

Food Delivery Management System

Purpose:

A Food Delivery Management System is used to manage online food ordering and delivery. It connects Customers, Restaurants, Food items, Orders, Payments, and Delivery partners in one database.

The database helps to:

- Store customer information
- Manage restaurants and their menus
- Record customer orders
- Track payments
- Assign delivery partners
- Maintain delivery status

REQUIRED TABLES

A practical Food Delivery Management System can be represented using 7 tables:

Customer
Restaurant
Menu_Item
Orders
Order_Item
Payment
Delivery_Partner

TABLES AND THEIR ATTRIBUTES

Table 1: CUSTOMER

Column	Data Type	Key
Customer_ID	INT	Primary Key
Customer_Name	VARCHAR	-
Phone	VARCHAR	-
Email	VARCHAR	-
Address	VARCHAR	-

Purpose:

Stores the details of customers who use the food delivery application.

Table 2: RESTAURANT

Column	Data Type	Key
Restaurant_ID	INT	Primary Key
Restaurant_Name	VARCHAR	-
Phone	VARCHAR	-
Rating	DECIMAL	-
Address	VARCHAR	-

Purpose:

Add a little bit of body text about restaurants available on the application

Table 3: MENU_ITEM

Column	Data Type	Key
Item_ID	INT	Primary Key
Restaurant_Name	VARCHAR	Foreign Key
Item_Name	VARCHAR	-
Category	VARCHAR	-
Price	DECIMAL	-
Availability	BOOLEAN	-

Purpose:

Stores the food items offered by each restaurant.

Relationship:

One restaurant can have many menu items.

Table 4: ORDERS

Column	Data Type	Key
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Restaurant_ID	INT	Foreign Key
Order_Date	DATE	-
Total_Amount	DECIMAL	-
Order_Status	VARCHAR	-

Purpose:

Stores information about orders placed by customers.

Relationship:

- A customer can place many orders.
- A restaurant can receive many orders.

Table 5: ORDER_ITEM

Column	Data Type	Key
Order_Item_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Item_ID	INT	Foreign Key
Quantity	INT	-
Subtotal	DECIMAL	-

Purpose:

Stores the individual food items included in each order..

Relationship:

- One order can contain many order items.
- One menu item can appear in many orders.

Table 6: PAYMENT

Column	Data Type	Key
Payment_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Payment_Date	DATE	-
Payment_Method	VARCHAR	-
Amount	DECIMAL	-
Payment_Status	VARCHAR	-

Purpose:

Records payment information associated with each order.

Relationship:

An order is associated with a payment record.

Table 7: DELIVERY_PARTNER

Column	Data Type	Key
Delivery_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Partner_Name	VARCHAR	-
Phone	VARCHAR	-
Delivery_Status	VARCHAR	-

Purpose:

Stores information about the delivery partner responsible for delivering an order.

Relationship:

A delivery partner can deliver multiple orders over time.

RELATIONS BETWEEN THE TABLES

Relationship	Cardinality	Meaning
Customer → Orders	1 : M	One customer can place many orders
Restaurant → Menu_Item	1 : M	One restaurant can have many menu items
Restaurant → Orders	1 : M	One restaurant can receive many orders
Orders → Orders_Item	1 : M	One order can contain multiple items
Menu_Item → Orders_Item	1 : M	One menu item can occur in many orders
Orders → Payment	1 : 1	Each order has one payment record
Orders → Delivery_Partner	1 : 1	An order is assigned to a delivery record